Yasra Chandio

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Department of Electrical and Computer Engineering
University of Massachusetts Amherst
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Education

PhD in Computer Engineering, University of Massachusetts Amherst. 2020–2026

MS in Computer Science, Binghamton University (SUNY). 2018–2019

BS in Software Engineering, Mehran University of Engineering and Technology, Pakistan *[Bronze]* 2010–2014

Awards Highlights

ML and Systems Rising Star | Link 2025

IEEE VR Honorable Mention For Best Paper | Top 3% Papers 2025

MIT EECS Rising Star | 19% acceptance rate | Link 2024
— *Highly selective workshop based on academic excellence and commitment to advancing DEI.*

Cyber-Physical Systems (CPS) Rising Star | 16% acceptance rate | Link 2024
— *Highly selective workshop based on outstanding achievements and future potential in CPS.*

UMASS College of Engineering Diversity, Equity, and Inclusion (DEI) Award | Link 2024
— *Award recognizing one undergraduate/graduate student/postdoc recognizing contributions to DEI.*

Research Featured in National & International News | BBC, UMASS, CoE, ScienceDaily, TechXplore 2024

Young Researcher at 10th Heidelberg Laureate Forum | 11% acceptance rate | Link 2023
— *A prestigious forum inviting 200 exceptional young researchers in mathematics and CS worldwide.*

Best Ph.D. Forum Presentation Award at ACM SenSys 2022

Computing Research Association Education **(CRA-E) Graduate Fellow** | Two-year term | Link 2022
— *National committee offers a graduate student role advocating undergraduate research.*

Three Minute Thesis (3MT) finalist at UMASS | Talk Link 2022

Publications

My work has been published at the top venues in virtual, augmented, and mixed reality (VR/AR/MR), robotics, systems, and machine learning, including IEEE VR, ISMAR, TVCG, NeurIPS, ICLR, ACM Sensys, ACM FAccT, IROS, and BuildSys/e-Energy. Complete publication list is available at https://scholar.google.com/citations?hl=en&user=LWUG9yIAAAAJ

Peer-reviewed

[1] **Yasra Chandio***, Diana Romero*, Salma Elmalaki, and Fatima M. Anwar: *What Sensors See, What People Feel: Exploring Subjective Collaboration Perception in Mixed Reality*, IEEE Conference on Virtual Reality and 3D User Interfaces (**VR**), 2026.

[2] Diana Romero*, **Yasra Chandio***, Fatima Anwar, and Salma Elmalaki: *MURMR: A Multimodal Sensing Pipeline for Automated Group Behavior Analysis in Mixed Reality*, IEEE International Symposium on Mixed and Augmented Reality (**ISMAR**), 2026. **[Honorable Mention Best Paper Award, Top 2% paper]** [1]

[*] *equal contribution*

[3] Momin Ahmad Khan, **Yasra Chandio**, and Fatima M. Anwar: *SABRE-FL: Selective and Accurate Backdoor Rejection for Federated Prompt Learning*, The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.

[4] M. Abdullah Soomro, M. Shayan Nazeer, Collin DelSignore, **Yasra Chandio**, M. Taqi Raza, and Fatima Anwar: *SynchroNB: Toward Robust Timing for 5G NB-IoT Networks.*, ACM/IEEE The International Conference on Embedded Artificial Intelligence and Sensing Systems (**Sensys**), 2026.

[5] John Murray*, **Yasra Chandio***, and Michael Zink: *Exploring the Relationship Between Quality of Experience and Quality of Service in Collaborative Mixed Reality.*, ACM International Conference on Interactive Media Experiences (**ACM IMX**), 2026.

[6] **Yasra Chandio**, Victoria Interrante, and Fatima Anwar: *Reaction Time as a Proxy for Presence in Mixed Reality with Distraction*, IEEE Conference on Virtual Reality and 3D User Interfaces (**VR**), 2025. **[Honorable Mention Best Paper Award, Top 3% paper]** [1]

[7] **Yasra Chandio**, Victoria Interrante, and Fatima Anwar: *Tap into Reality: Understanding the Impact of Interactions on Presence and Reaction Time in Mixed Reality*, IEEE Conference on Virtual Reality and 3D User Interfaces (**VR**), 2025. [1]

[8] **Yasra Chandio**, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar: *Lost in Tracking Translation: A Comprehensive Analysis of Visual SLAM in Human-Centered XR and IoT Ecosystems*, ACM Transactions on Sensor Networks (**TOSN**), 2025. **[AIModels.fyi feature link]**

[9] Momin Ahmad Khan, Virat Shejwalkar, **Yasra Chandio**, Amir Houmansadr, and Fatima M. Anwar: *Decoding FL Defenses: Systemization, Pitfalls, and Remedies*, ACM Transactions on Privacy and Security (**TOPS**), 2025.

[10] Abdullah Al Ishtiaq*, Raja Hasnain Anwar*, **Yasra Chandio**, Fatima M. Anwar, Syed Rafiul Hussain, and Muhammad Taqi Raza: *Cloud Nine Connectivity: Evading in-Flight Paywalls For Free Internet*, ACM Conference on Security and Privacy in Wireless and Mobile Networks (**ACM WiSec**), 2025.

[11] **Yasra Chandio**, Victoria Interrante, and Fatima Anwar: *Human Factors At Play: Understanding The Impact Of Conditioning On Presence And Reaction Time In Mixed Reality*, IEEE Conference on Virtual Reality and 3D User Interfaces (**VR**), 2024. [1] **[Top 10% paper]**

[12] **Yasra Chandio**, Momin Ahmad Khan, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima Anwar: *A Neurosymbolic Approach To Adaptive Feature Extraction In SLAM*, IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024. **[5-minute Video Teaser]**

[13] **Yasra Chandio**, Noman Bashir, and Fatima Anwar: *Stealthy and practical multi-modal attacks on mixed reality tracking*, IEEE International Conference on Artificial Intelligence & extended and Virtual Reality (**AIxVR**), 2024. **[Best Paper Finalist]**

[14] Momin Ahmad Khan, **Yasra Chandio**, and Fatima Anwar: *HYDRA-FL: Hybrid Knowledge Distillation for Robust and Accurate Federated Learning*, Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2024.

[15] Khotso Selialia, **Yasra Chandio**, and Fatima Anwar: *Mitigating feature heterogeneity biases in Federated Learning*, ACM Conference on Fairness, Accountability, and Transparency (**FAccT**), 2024.

[16] **Yasra Chandio**, Noman Bashir, Tian Guo, Elsa Olivetti, and Fatima M. Anwar: *Scoping The Decarbonization Of Collaborative Mixed Reality*, IEEE International Symposium on Emerging Metaverse (**ISEMV**), 2024.

[17] **Yasra Chandio**, Noman Bashir, Victoria Interrante, and Fatima Anwar: *Investigating the Correlation Between Presence and Reaction Time in Mixed Reality*, IEEE Transactions on Visualization and Computer Graphics (**TVCG**), 2023. **[Selected for Presentation at IEEE VR 2024]**

[1] Due to substantial contribution selected to appear in TVCG Special Issue.

[18] Noman Bashir, **Yasra Chandio**, David Irwin, Fatima Anwar, Jeremy Gummeson, and Prashant Shenoy: *Jointly managing electrical and thermal energy in solar- and battery-powered computer systems*, ACM International Conference on Future Energy Systems (**e-Energy)**, 2023.

[19] **Yasra Chandio**, Aditya Mishra, and Anand Seetharam: *GridPeaks: Employing Distributed Energy Storage For Grid peak Reduction*, IEEE International Green and Sustainable Computing conference (**IGSC)**, 2019.

[20] **Yasra Chandio**, Jo A gila Bitsch, Affan A. Syed, and Muhammad Hamad Alizai: *Networking Wireless Energy in Embedded Networks*, ACM Transactions on Sensor Networks (**TOSN)**, 2018.

[21] Samar Abbas, Abu Bakar, **Yasra Chandio**, Khadija Hafeez, Ayesha Ali, Tariq M Jadoon, and Muhammad Hamad Alizai: *Inverted HVAC: Greenifying Older Buildings, One Room at a Time*, ACM Transactions on Sensor Networks (**TOSN)**, 2018.

[22] Khadija Hafeez, **Yasra Chandio**, Abu Bakar, Ayesha Ali, Affan A. Syed, Tariq M. Jadoon, and Muhammad Hamad Alizai: *Inverting HVAC for Energy Efficient Thermal Comfort in Populous Emerging Countries*, ACM International Conference on Systems for Energy-Efficient Built Environments (**BuildSys)**, 2017. **[Audience Choice Award]**

[23] M. Hamad Alizai, Qasim Raza, **Yasra Chandio**, Affan A. Syed, and Tariq M. Jadoon: *Simulating Intermittently Powered Embedded Networks*, 13th International Conference on Embedded Wireless Systems and Networks (**EWSN)**, 2016.

Selected Work In Progress

[1] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Discovering Causal Pathways to Presence in Mixed Reality with Latent-Aware Models.*

[2] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar: *LipFed: Mitigating Subgroup Bias in Federated Learning with Lipschitz Constraints*. **[Preprint]**

[3] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar: *ADAPT-FED: Adaptive Federated Optimization with Learning Stability*. **[Preprint]**

Workshop/Posters/Late-Breaking Work

[1] Momin Ahmad Khan, **Yasra Chandio**, Eugene Bagdasarian, and Fatima Anwar: *Poster: Compromising Federated Medical AI-Backdoor Risks in Prompt Learning*, ACM Conference on Embedded Networked Sensor Systems (SenSys), 2025.

[2] Wacuka M Ngata, Noman Bashir, Michelle Westerlaken, Laurent Liote, **Yasra Chandio**, and Elsa Olivetti: *The Cloud Next Door: Investigating the Environmental and Socioeconomic Strain of Datacenters on Local Communities*, ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies - Late-Breaking Work (ACM COMPASS), 2025.

[3] **Yasra Chandio**, Victoria Interrante, and Fatima Anwar: *Balancing Presence And Safety Using Reaction Time In Mixed Reality*, Workshop at IEEE International Symposium on Mixed and Augmented Reality (SafeAR@ISMAR), 2024.

[4] **Yasra Chandio**, Noman Bashir, and Fatima Anwar: *Dataset: HoloSet - A dataset for visual inertial pose tracking in extended reality*, 5th International SenSys/BuildSys Workshop on Data (DATA), 2022.

[5] **Yasra Chandio**: *Towards secure and immersive mixed reality: PhD Forum Abstract*, 20th ACM Conference on Embedded Networked Sensor Systems (SenSys), 2022. **[Best Presentation Award]**

[6] Khotso Selialia, **Yasra Chandio**, and Fatima Anwar: *Federated Learning Biases in Heterogeneous Edge-Devices - A Case-Study*, 4th International SenSys/BuildSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengeIoT), 2022.

[7] **Yasra Chandio** and Fatima Anwar: *Poster: Spatiotemporal Security in Mixed Reality systems*, 18th ACM Conference on Embedded Networked Sensor Systems (SenSys), 2020.

Open Source Research Artifact

HoloSet - A Dataset for Visual-Inertial Pose Estimation in Extended Reality | Link

Invited Talks

Sensing, Interpreting, and Adapting to Human Experience 2026

— *Learning Engineering and Practice Group (LEAP Group), Massachusetts Institute of Technology.*

— *Department of Computer Science, University of Texas at Dallas.*

System Signals to Felt Experience: Designing Human-Centered Mixed Reality 2025

— *Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign.*

— *Department of Electrical and Computer Engineering (I3T Lab), Duke University.*

— *Department of Information Science and Technology, George Mason University.*

— *IMMERSE Symposium at University of Illinois Urbana-Champaign.*

— *MLSys Rising Star Workshop, Meta HQ.*

— *AT&T Labs.*

— *Mitsubishi Electric Research Laboratories.*

A Human-Centered Approach to Immersive and Adaptive Mixed Reality 2025

— *Nokia Bell Labs.*

Towards Secure and Immersive Mixed Reality 2023

— *Heidelberg Laureate Forum*

— *ACM Sensys PhD Forum*

Media Coverage

Can AR and VR finally disrupt the exhausting culture of video meetings?	BBC
ECE's Yasra Chandio Qualifies as a "Rising Star" in Both Actual and Virtual Reality	CoE
College of Engineering Selects LaChance and Chandio for Its Diversity, Equity, and Inclusion Award	CoE
CRA Education Committee Selects New Graduate Fellow	CRA
New Study: Immersive Engagement In Mixed Reality Can Be Measured With Reaction Time	UMASS
Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time	ScienceDaily
Study: Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time	TechXplore

Awards and Honors

Nokia Bell Labs Gift for Society for Women Researchers | $2000 2024

UMASS Campus Climate Improvement Grant | $2500 2024

Scholar at CRA-WP Grad Cohort Workshop for IDEALS 2024

IEEE Robotics and Automation Society (RAS) travel grant to attend IROS 2024

IEEE VR DEIA Award to attend IEEE VR 2024

Abbe Grant to attend HLF | Highly selective grant, given to 30/200 HLF attendees 2023

CPS-IoT Week SIGBED student travel award | declined 2023

Scholar at CRA-WP Grad Cohort 2022

Invited to attend Google PhD Fellowship Summit 2021

Honorable mention in UMASS ECE 3MT 2021

Selected for Computer Research Mentorship (CSRMP) Program at Google 2021

Grace Hopper Scholar to attend Grace Hopper Celebration 2019

National Science Foundation travel grant to attend IEEE IGSC 2019

Summer Provost Fellowship at Binghamton University 2019

Best Paper Audience Choice Award at ACM BuildSys 2017

ACM SenSys Travel grant to attend ACM BuildSys/SenSys 2017

Service

Leadership

Founder and Chair of multiple initiatives advancing belonging and inclusion in engineering 2023–2026

— *College of Engineering Dean's Graduate Advisory Group.*

— *ECE Grad Gender Diversity group.*

— *UMASS ECE Women Dinner Series | UMASS Campus Climate Grant ($2500) | [Link]*

— *ECE Grad Women Retreats | Raised $2500*

Co-organizer, UMASS CoE GradLife Event Series | Research Symposium 2024–2026

Organizer and Panelist, ECE Graduate Orientation 2023–2025

Inaugural Graduate Student Inductee in ECE DEI Committee 2023–2026

Editor, CRA-Undergraduate Research Highlights | [Link] 2022–2024

— *The series features outstanding undergraduate research across North America, offering insights into research experiences, mentorship, career advice, and promoting graduate education.*

Service to the Profession

Technical Program Committee Chair for in-progress work and Demo IEEE AIXVR 2026

Technical Program Committee member at: 2026

—IEEE VR, IEEE ISMAR, VRST, ACM MMSys, ACM Multimedia, IEEE ICCCN, XRSecurity@Mobicom

Associate Chair (AC) at ACM IMX 2026

General Chair at DataFM@BuildSys 2025

Technical Program Committee Chair at Sensors S&P@CPSIOT-Week 2025

Publications Chair at SmartSP 2025

Associate Chair (AC) at ACM IMX 2025

Technical Program Committee member at ACM VRST, XRsecurity@MobiHoc, ACM Sensys Posters 2025

Reviewer at ICLR, IEEE VR, ACM CHI, IEEE ICRA, IMWUT, ACM MM 2025

Technical Program Committee member at DATA@Buildsys, ACM S3@Mobicom 2024

Reviewer at Journal of Parallel and Distributed Computing (JPDC) 2024

Shadow Program Committee member at ACM SenSys and ACM EuroSys 2022

Review Committee member at Extended Reality track at Grace Hopper Celebration (GHC) 2022

Student Volunteer at ACM SenSys 2022

Review Committee member at VR/AR/MR track at GHC 2021

Teaching and Mentorship

`Instructor`

ECE 231: Introduction to Embedded System — `Summer'25`

Mixed Reality, Programming, and Capstone Projects (UMASS Turing Program) — `2022–2025`

Embedded System Programming (UMASS 3-week K-12 Engineering Outreach Summer Program) — `2021`

`Teaching Assistant`

ECE 231: Introduction to Embedded System – helped design new labs — `Spring 21–25`

ECE 524/624: Advanced Embedded Systems (Graduate) – helped design course projects — `Fall 21–23`

ECE 322: Systems Programming — `Fall 24–25`

ECE 571 (SUNY): Programming Languages (Graduate) — `Fall 19`

CS 677 (LUMS): Internet of Things — `Fall 16`

`Guest Lecturer`

Mixed Reality Beyond Pokemon Go (Pioneer Valley Chinese Immersion Charter School) — `2024`

Neurosymbolic Human Pose Tracking (CS+Robotics Club at Nashua High School, New Hampshire) — `2024`

Deep Learning based Tracking for Mobile Systems (UMASS ECE 524/624) — `2023`

Mixed Reality Security and Privacy (UMASS ECE 524/624) — `2022`

Introduction to Embedded System with *BeagleBone Black* (UMASS ECE 231) — `2020`

`Panelist`

Ask a Grad Student: Real Advice for Navigating Your Journey (UMASS CoE) — `2025`

Research Career and Graduate School (UMASS ECE) — `2024`

— *A panel discussion on academic careers and life at graduate school.*

Surviving PhD (UMASS Graduate Women In STEM) — `2024`

— *A panel discussion connecting UMASS graduate and undergraduate students to discuss careers in STEM.*

Organizer & Moderator, Life at UMASS as a CS Undergraduate Student (UMASS Turing Program) — `2023`

Organizer & Moderator, UMASS Women in Engineering Day — `2023`

— *A panel discussion connecting industry with high school students to discuss careers in engineering.*

Pathways To Graduate Research (UMASS ECE) — `2023`

`Mentorship`

Eli Rotondo (Duke PhD) — `2026–`

John Murray (UMASS PhD) — `2024–`

Diana Romero (UC Irvine PhD) — `2023–`

Standly Laguerre | NSF REU (Macalester College), now Software Engineer at an AI startup. — `2023`

*UMASS **Undergrads** (2021 - 2023):* (1) Kenneth Lin; (2) Yav Rohatgi; (3) Colin Leydon; (4) Eyado Ocho; (5) John Dale; (6) Dani Kasti.

*UMASS **Graduate Students** (2020 - 2023):*
(1) Alexander Dickopf; (2) Linghao Meng; (3) Zhehang Zhang ; (4) Yanan Zhang; (5) Zachary Wannie; (6) Yifan Meng; (7) Xin Zhao; (8) Subhankar Chowdhury; (9) Abhinaba Das; (10) Xiaohao Xia; (11) Yinxuan Wu; (12) Xintao Ding; (13) Shaoyu Dai ; (14) Nikhil Sarecha; (15) Rehmat Singh.

Employment

Graduate Research Assistant, UMASS Amherst | *Advisor: Fatima Anwar* `2020–on`

Graduate Research Assistant, Binghamton University | *Advisor: Gaunhua Yan* `2018–2019`

Research Associate, LUMS, Pakistan | *Advisor: Muhammad Hamad Alizai* `2015–2017`

Research Engineer, NUCES, Pakistan | Advisor: *Affan A. Syed* `2014–2015`

References

1. **Fatima Anwar**, Assistant Professor at University of Massachusetts Amherst fanwar@umass.edu
2. **Victoria Interrante**, Professor at University of Minnesota, Twin Cities interran@umn.edu
3. **Michael Zink**, Professor at University of Massachusetts Amherst mzink@umass.edu
4. **Luis Antonio Garcia**, Assistant Professor at University of Utah la.garcia@utah.edu
5. **Joseph DeGol**, Principal AI Research Lead, Microsoft joe.degol@outlook.com